



Central Limit Theorem

Why averages often become normally distributed

Simple explanations • worked examples • practical use cases



1. The central idea

Take many random samples of the same size from a population and calculate the mean of each sample. As sample size becomes sufficiently large, the distribution of those sample means becomes approximately normal, even when the original population is not normal.

The theorem concerns the distribution of sample means, not necessarily the distribution of individual observations.

Three important results

- The mean of sample means is approximately the population mean.
- The distribution of sample means becomes approximately bell-shaped.
- Larger samples produce less variation among sample means.

2. Standard error

The standard deviation of sample means is called the standard error.

$$\text{Standard error of the mean} = \sigma / \sqrt{n}$$

If population standard deviation $\sigma = 100$ and sample size $n = 25$, standard error = $100/5 = 20$. If n increases to 100, standard error becomes $100/10 = 10$.

To cut the standard error in half, the sample size must be multiplied by four.

3. Banking example

Average transaction value

Individual transaction amounts are usually right-skewed because many are small and a few are very large. A bank repeatedly draws random samples of 50 transactions and calculates each sample mean.

- The individual amounts may remain strongly skewed.
- The sample means will usually appear more bell-shaped.
- Most sample means will cluster near the true population mean.
- The spread of sample means is measured by the standard error.

This allows the bank to construct confidence intervals and perform hypothesis tests about average transaction values.

4. Conditions and misconceptions

Practical conditions

- Samples should be random and observations reasonably independent.
- A sample size of 30 is only a rough guideline, not a universal rule.
- Strong skewness or extreme outliers may require much larger samples.
- For sampling without replacement, the sample should generally be less than 10% of the population.

What it does not mean

- It does not say every dataset becomes normal.
- It does not say the population becomes normal.
- It does not guarantee that $n = 30$ is always sufficient.
- It does not remove sampling bias or poor data quality.